Q²-JEPA

Dual-Quantum Visual World Model Architecture

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CUDA-Q Accelerated visual world model uniting Quantum Swap-Test Self-Attention and Born-Machine Latent Priors

Motivation

LLMs predict words. World models predict the world.

LLM

"The car is approaching the..."

Text in

"...intersection"

Next word out

World model



Scene now



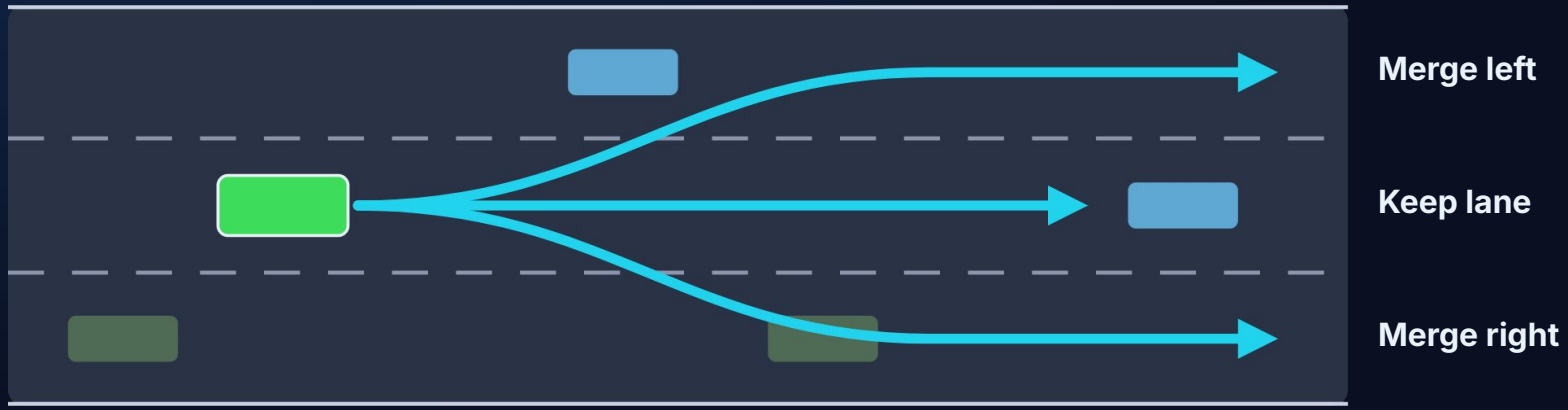
Predicted next state of the world

Used for

- Self-driving
- Robotics
- Planning
- Simulation

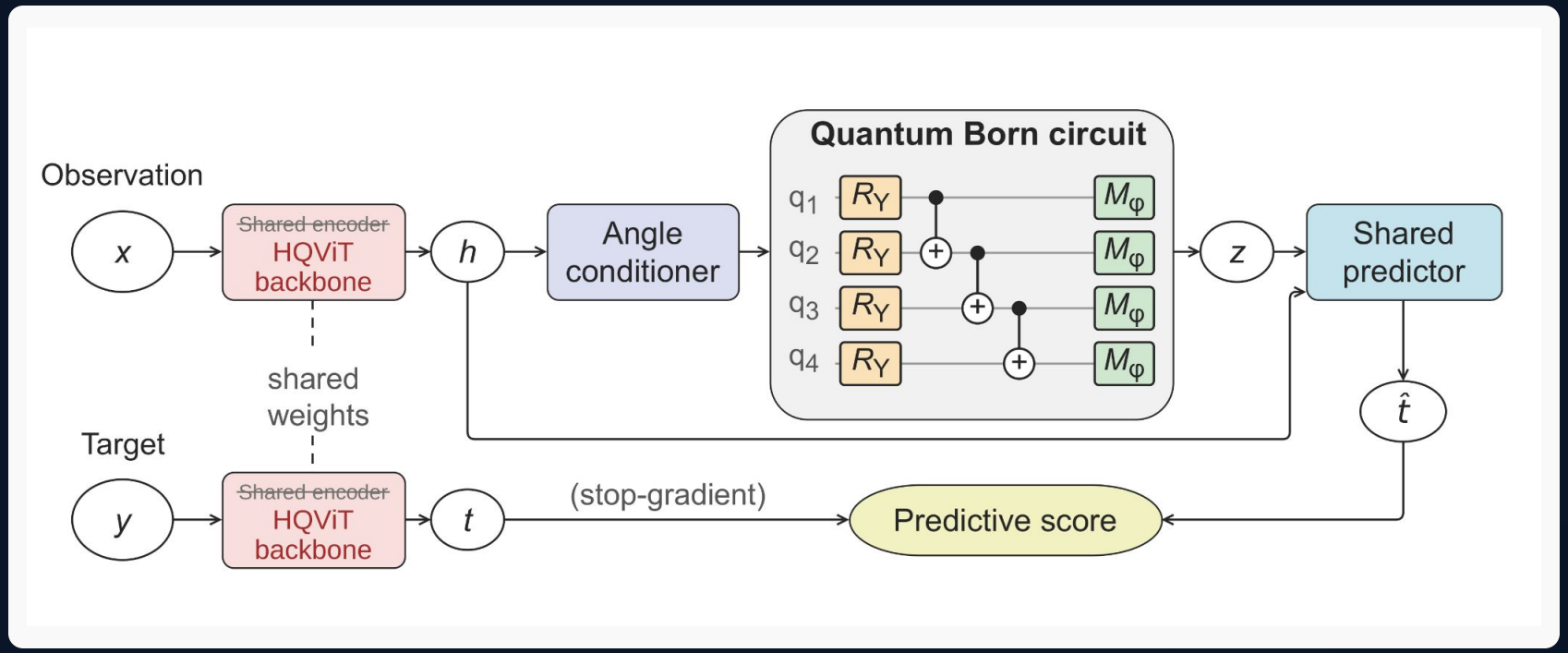
One moment, many futures.

Averaging them gives the middle path. That's why Q²-JEPA samples each one.



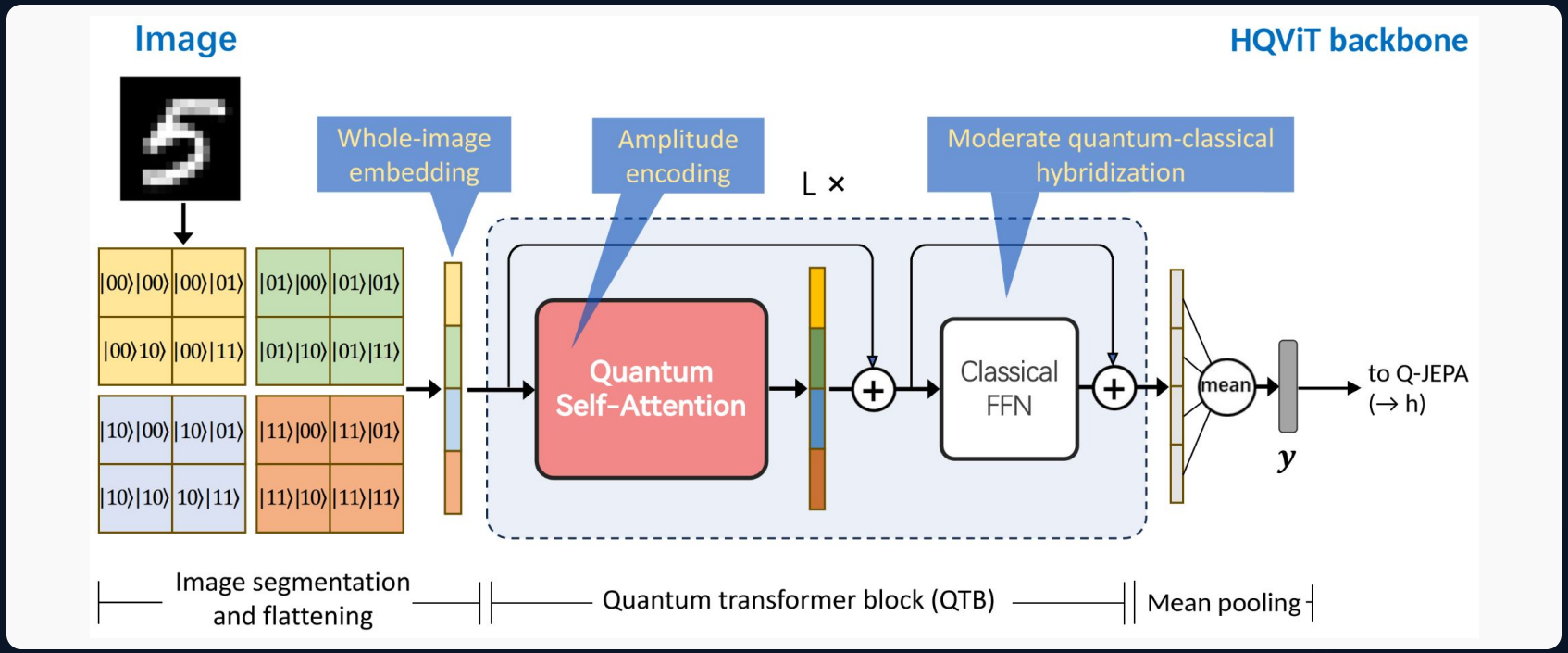
SYSTEM ARCHITECTURE

Q²-JEPA System Workflow — Quantum Born Priors for Latent Prediction



ENCODER BACKBONE

HQViT Backbone Architecture — Hybrid Quantum Vision Transformer Block



WHY CUDA-Q?

Why CUDA-Q? — CUDA powers AI on GPUs. CUDA-Q powers Quantum + AI together.

CPU Quantum

Simulate 17 qubits + 1B shots on CPU...
Separate CPU & GPU memory

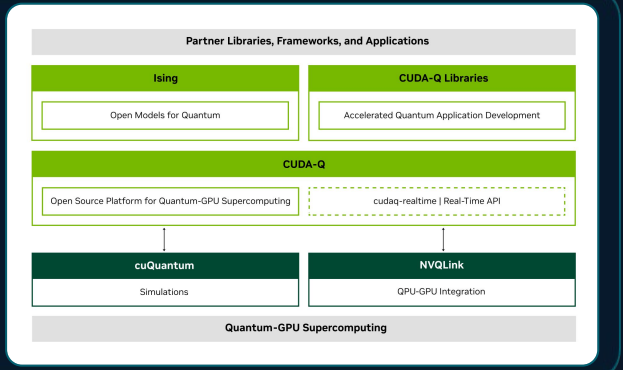
→

...Weeks of training
Bottlenecked pipeline

NVIDIA CUDA-Q

Unified GPU + QPU execution
Direct high-bandwidth memory interconnect with noise-aware compilation

→



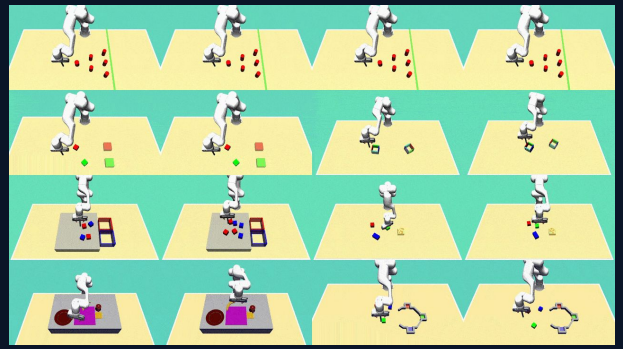
Used for

- Quantum Machine Learning
- Multi-Agent Coordination
- Robotic Planning
- Sim-to-QPU Hardware

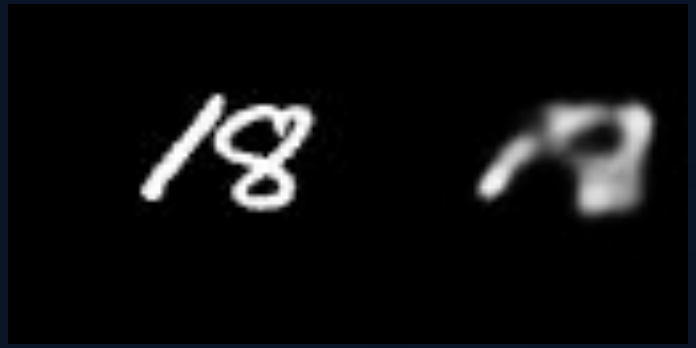
EXPERIMENTAL EVALUATION

Benchmark Environments

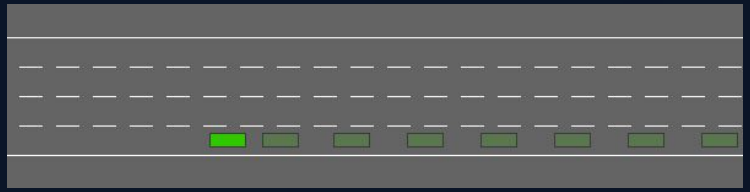
D3IL Robotics Benchmark [MANIPULATION]



Moving MNIST [SPATIOTEMPORAL]



Highway-Env Simulation [AUTONOMOUS DRIVING]



LITERATURE & FOUNDATIONS

Key References — Foundational Quantum Computing & World Model Publications

01

QUANTUM WORLD MODELS

Z. Liu, S. Liu, H. Wu. *"Quantum-JEPA: Quantum Born Priors for Latent-Action JEPA World Models."* arXiv:2610.27921, 2026.

02

HYBRID QUANTUM VISION

H. Zhang, Q. Zhao, M. Zhou, L. Feng. *"HQViT: Hybrid Quantum Vision Transformer for Image Classification."* arXiv:2504.02730, 2025.

03

QUANTUM COMPLIANCE & SDK

NVIDIA Quantum Computing Platform. *"CUDA-Q: Accelerated Quantum-Classical Computing Architecture."* nvidia.github.io/cuda-quantum